

**Please solve the following exercises at home prior to the tutorial**

### Exercise 1

Given is an undirected simple graph defined by a set of vertices  $V$  and a set of edges  $E$ :

$V = \{a, b, c, d, e, f, g\}$

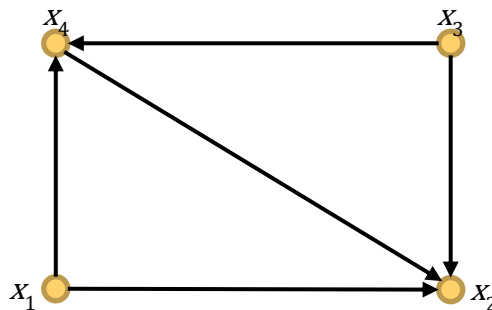
$E = \{ \{a, b\}, \{b, c\}, \{a, c\}, \{a, e\}, \{f, g\} \}$

- Draw the graph.
- How many connected components does this graph have? Name them. Are there separating vertices?
- Give the degree of each vertex.
- Does the graph contain cycles?

**The following exercises will be done during the tutorial:**

### Exercise 2

Given is the following directed graph:



- Determine the adjacency matrix  $A$  and its powers  $A^r$  ( $r = 2, 3, 4$ ).
- What does this tell us about possible walks and cycles?